

MINOR PROJECT ASSESSMENT

Networking, Linux & Reconnaissance

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Project	Mini Reconnaissance Project – Q5
Scope	Authorized lab environment and assignment-specified example.com reconnaissance

1. Project Objective

The objective of this mini project was to perform structured reconnaissance using networking and Linux commands, browser-based technology discovery, web directory enumeration, and passive subdomain discovery. The activities were documented with commands, observed results, and short security-focused analysis.

Authorization note: Testing was limited to the authorized lab target used for web enumeration and the assignment-specified example.com domain. No exploitation was performed.

2. Phase 1 – Network & Connectivity

Kali Linux networking was verified before reconnaissance. The final configuration showed the Kali interface using **10.0.2.15/24**, with default gateway **10.0.2.2**. Gateway connectivity was confirmed with 0% packet loss. DNS configuration showed Google Public DNS servers 8.8.8.8 and 8.8.4.4.

```
ip addr
ip route
ping -c 4 10.0.2.2
cat /etc/resolv.conf
```

```
(kali@kali)-[~]
$ ping -c 4 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data:
64 bytes from 10.0.2.2: icmp_seq=1 ttl=64 time=0.506 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=64 time=0.698 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=64 time=0.207 ms
64 bytes from 10.0.2.2: icmp_seq=4 ttl=64 time=0.326 ms

--- 10.0.2.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3076ms
rtt min/avg/max/mdev = 0.207/0.434/0.698/0.185 ms

(kali@kali)-[~]
$ cat /etc/resolv.conf
# Generated by NetworkManager
search lan
nameserver 8.8.8.8
nameserver 8.8.4.4
nameserver fd17:625c:f037:2::3
```

Figure 1. Final gateway connectivity and DNS configuration evidence.

3. Phase 2 – Technology Discovery

Wappalyzer identified **Cloudflare** as a CDN for example.com. This is useful during authorized reconnaissance because it indicates that a CDN/proxy layer is involved in delivering website content and helps build an understanding of the site's infrastructure.

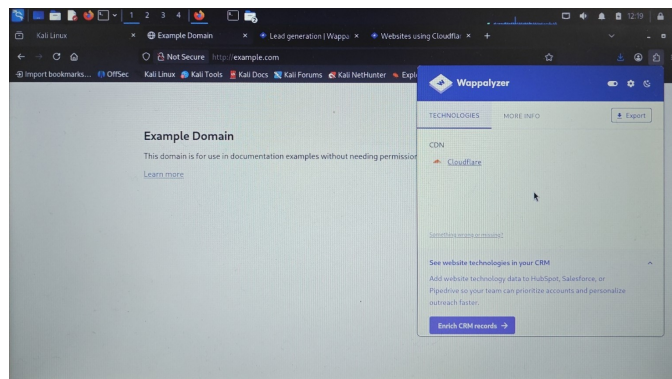


Figure 2. Wappalyzer technology detection for example.com.

4. Phase 2 – Netcraft Findings

The Netcraft Site Report provided infrastructure and technology information for example.com. The observed report identified **Cloudflare, Inc.** as the hosting company/netblock owner, hosting country as the **United States**, IPv4 address **172.66.147.243**, and Cloudflare-related site technology. It also showed Gzip HTTP compression and HTML5 document type. These observations support infrastructure mapping without exploitation.

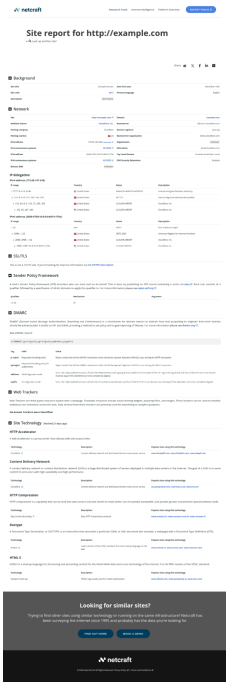


Figure 3. Netcraft Site Report for example.com.

5. Phase 3 – Web Enumeration with Gobuster

An authorized Metasploitable3 lab web server at **172.28.128.3** was enumerated using Gobuster directory mode and the common Dirb wordlist.

```
gobuster dir -u http://172.28.128.3/ -w /usr/share/wordlists/dirb/common.txt
```

Path	Status	Interpretation
/chat/	301	Redirected directory; requires further authorized review.
/drupal/	301	Redirected application path; useful for application mapping.
/phpmyadmin/	301	Redirected management/application path; requires authorized review.
/uploads/	301	Redirected upload-related path; useful for understanding site structure.
.htpasswd	403	Forbidden response; access is restricted.
.htaccess	403	Forbidden response; access is restricted.
cgi-bin/	403	Forbidden response; access is restricted.

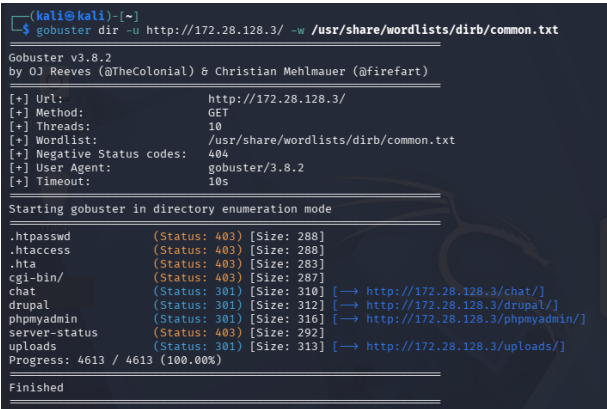


Figure 4. Gobuster directory enumeration output. The scan completed 4613/4613 requests (100%).

6. Phase 4 – Passive Subdomain Enumeration

Subfinder was used for passive subdomain discovery against the assignment-specified domain. Because passive sources can return noisy or unexpected entries, DNS validation was used before treating a result as resolvable.

```
subfinder -d example.com -silent
after discovery:
subfinder -d example.com -silent | dnsx -silent
```

The DNSx validation confirmed **www.example.com** as a DNS-resolvable domain. This demonstrates the value of separating discovered names from those that can actually be resolved in DNS.



Figure 5. DNSx validation showing www.example.com.

7. Phase 5 – Analysis & Security Relevance

Area	Key observation	Why it matters
Network	Kali IP 10.0.2.15/24; gateway 10.0.2.2 reachable.	Confirms the lab machine has working network connectivity before testing.
Technology	Cloudflare identified by Wappalyzer/Netcraft.	Helps map the site's delivery/infrastructure layer during authorized reconnaissance.
Web paths	Several 301 and 403 responses found by Gobuster.	Reveals application structure and access-controlled resources for further authorized review.
Subdomains	www.example.com resolved through DNSx.	DNS validation reduces reliance on unverified passive enumeration results.

8. Shodan Note

A generic Shodan search for example.com did not provide a verified target-specific result for this report. The hostname filter required login, so no Shodan finding was fabricated or treated as confirmed. This is documented as a limitation rather than bypassing the access requirement.

9. Conclusion

The reconnaissance exercise demonstrated a complete workflow from network verification to technology discovery, web enumeration, and DNS-validated subdomain discovery. The results were interpreted using HTTP status codes and DNS resolution rather than attempting exploitation. The exercise also showed why reconnaissance results should be validated before being used for further authorized security testing.

Submission checklist: Commands included ✓ Relevant screenshots included ✓ Findings explained ✓ Conclusion included ✓